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LMSRE-Based Adaptive PI Controller for Enhancing the Performance of an Autonomous Operation of Microgrids

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ABSTRACT Microgrids take a large part in power networks thanks to their operational and economic benefits. This research introduces a novel implementation of an adaptive proportional plus integral (PI) controller to boost the autonomous microgrid operation efficiency. The least mean and square roots of the exponential algorithm are utilized in the adaptive PI control strategy. The multi-objective function for both sunflower optimization (SFO) and particle swarm optimization (PSO) algorithms is obtained by The Response Surface Methodology. The system is evaluated under different environments, which are stated as follows: 1) disconnect the system from the grid (islanding), 2) autonomous system exposure to load variability, and 3) autonomous system exposure to a symmetrical fault. The proposed practicality of the control plan is shown by the data of the simulation, which is extracted from PSCAD/EMTDC software. The strength of the suggested adaptive control is confirmed through matching its results with those obtained using the SFO and PSO based optimal PI controllers.

INDEX TERMS Microgrid, optimization, power systems, renewable energy.

I. INTRODUCTION

A. LITERATURE REVIEW

The large centralized generation stations continue struggling to satisfy the growth in demand. That's why the trend in generation nowadays is looking towards the smaller distributed generation (DG) networks, situated near to the load centers. This minimizes the transmission losses and stops the rapid expansion of the transmission network. The value of the DG systems has grown thanks to their assistance in improving the system stability, power efficiency, voltage profile, climate, and economic advantages. In specific, the DG is capable of:

- 1) Reducing greenhouse emissions to protect the atmosphere [1].
- 2) Giving easy solutions to catch the growth of load.
- 3) Stopping the rapid expansion of the transmission network which minimizes the transmission and distribution losses.

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Furthermore, the load efficiency and power quality would increase if the DGs are properly regulated [2].

The idea of microgrid (MG) is shined up due to the rapid increase of the DG penetration in the power system. The MG contains multiple DGs and different loads. It can work either grid linked or in an autonomous operation mode [3]. In the first mode, the MG is linked to the grid. In addition, the power balance between the load and the microgrid generation is not needed because any shortfall in the power will be supplied from the grid. Also, any extra power will be drained by the grid. Meanwhile, in the second mode, the microgrid is separated from the grid. In addition, the DG has to preserve the equilibrium of generation – demand. Moreover, the DG regulates the voltage and frequency in this mode of operation to keep them within limits.

The advanced control methods are utilized in autonomous mode to ensure robust and safe operation. These methods can be classified into three major groups: (1) droop-based, (2) centralized control, and (3) multivariable and servomechanism (MVAS) techniques. In [4], the droop-based method

is used to adjust the current of the DG units in a microgrid during the stand-alone mode. The benefits of using the droop-based control are its ability to independently control distributed energy resource units (DERs) without contact between them. Its principal downside is, however, not being able to deal with nonlinear systems like load dynamics, which affect the system transient response [5]. In comparison, centralized control strategies require high bandwidth connections, and any loss of such connections can lead to a breakdown in the microgrid [6]. Finally, MVAS controllers are utilized in DG for autonomous mode [7]. Though, its inconvenience is the high complexity. Meanwhile, the PI controller is the best. It has always led among the others in the nonlinear system due to its high margin of stability. Owing to the responsive impact of such parameter, tuning the PI controller faces great difficulties. Therefore, many meta-heuristic algorithms have been improved in order to overcome those difficulties, such as particle swarm optimization (PSO) [8], sunflower optimization algorithm (SFO) [9], [10], hybrid GWO-PSO optimization technique [11], genetic algorithm (GA) [12], hybrid firefly and particle swarm optimization technique [13], Harris hawks optimization Method [14], marine predators algorithm [15], hierarchical model predictive control [16], Tabu search [17], quasi-oppositional selfish herd optimization (QSHO) [18], Cuttlefish optimization algorithm (CFA) [19], and teaching-learning based optimization [20]. Each of those techniques has its benefits and drawbacks [21].

B. RESEARCH GAP AND MOTIVATION

The optimization approaches have many limitations, such as dynamic systems, high storage demands, and high microprocessor capacities, etc. These drawbacks mean that new control procedures, including adaptation techniques, are needed. In [22], adaptive control based on an affine projection algorithm is presented. Faster integration and less machine complexity than optimization strategies were demonstrated. The previous adaptive technique also demonstrated certain attractive stuffs such as the coefficient vector. This technique monotonously approaches the real data regardless of the inputs, as stated in [23]. In [24], a new adaptive control based on set-membership engine algorithm is provided with improved performance than other techniques. The continuous p-norm mixed technique is presented as a new adaptive control of the VS, which rapidly updates the fuzzy inference system in [25]. In [26], the improved multiband-structured subband adaptive filter (IMSAF) algorithm-based self-tuned PI controller is employed to adjust the interface voltage source converters through a cascaded control structure. In [27], the improved zero attracting quaternion-valued least mean square (IZAQLMS) control approach with high DC offset rejection capability is used to improve the power quality of the grid current and to ensure the unity power factor operation of the grid.

The authors have proven the pioneer performance of the adaptive control algorithms over the conventional algorithms,

SFO and PSO, in terms of less-computational complexity, better convergence, and lowest mean square error. Moreover, a class of reinforcement learning algorithms is not applicable for simple problem's solution as it requires huge data and lots of calculations.

This paper suggests a new adaptive technique titled as the least mean (LM) and square root of exponential (SRE) technique. The normalization of this technique is induced by the use of a negative absolute error exponent between [0, 1]. Furthermore, LMSRE has constants (μ and α) that combine rapid convergence and stability with minimal error.

C. CONTRIBUTION AND PAPER ORGANIZATION

The major contributions of this paper are:

- 1) Evaluation of the newly proposed LMSRE algorithm when applied to adapt PI controllers to enhance the MG system performance,
- 2) Testing the reliability of the proposed technique by exposing the system to several different environments, which are i) disconnect the system from the grid (islanding), ii) autonomous system exposure to load variability, and iii) autonomous system exposure to symmetrical faults,
- 3) Checking the validity of the presented adaptive control through matching its results with those obtained using the SFO and PSO algorithms.

The remaining sections of the paper are organized as follows: Section 2 presents the system modeling. Section 3 presents the control strategy. Section 4 presents the modeling stage using Response Surface Methodology (RSM), SFO and the suggested adaptive PI controller based on LMSRE technique. Section 5 presents the simulation results and discussion. Finally, the conclusion is presented in Section 6.

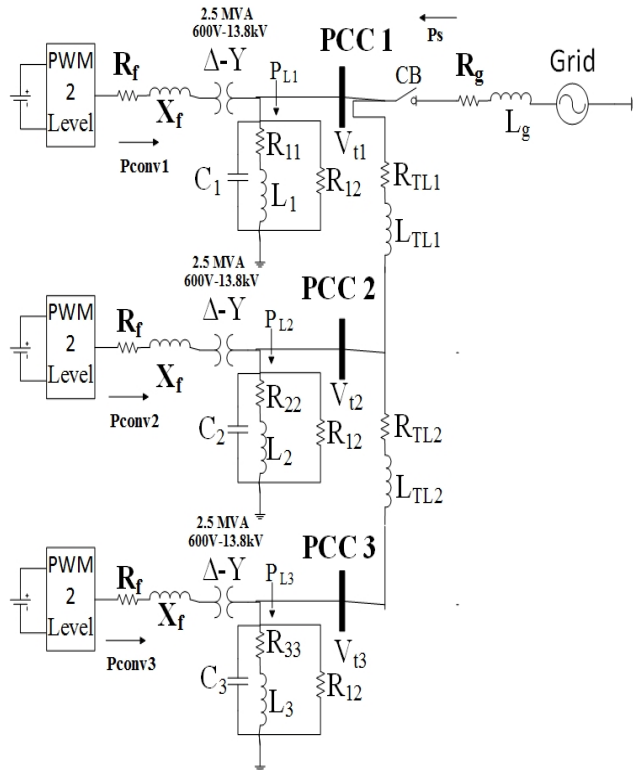
II. SYSTEM MODELLING

Fig.1 presents a single line diagram (SLD) of an MG, which consists of 3 DGs linked with each other by transmission lines. The DGs are linked to the PCC, which is linked to the grid by a transmission line (TL). Each DG consists of DC source linked to pulse width modulation (PWM) (2 levels). The PWM is linked to a Δ - Y transformer by a series filter to achieve a better quality of output power. A three-stage complex RLC load is attached to the transformer to reflect the local load. The MG data are stated in Table 1.

The system is working either in grid linked or in autonomous operation mode. In the grid linked mode, the DG unit runs in active and reactive power control mode. It is important to remember that the voltage and frequency are maintained by the grid. On the other hand, in the autonomous operation mode, the DG has to preserve the equilibrium of generation – demand. Also, it regulates the voltage and frequency to keep them within limits. This research focuses on enhancing the MG during the autonomous operation mode using the cascaded control scheme, which is discussed in the next section.

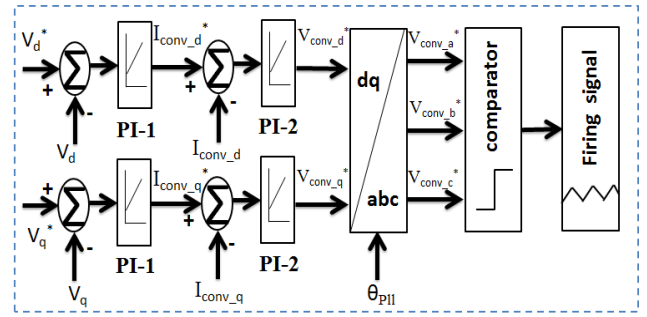
TABLE 1. System data for MG.

Transformer parameters	
$\Delta/Y = 0.6/13.8$ KV	
Load data	
Load 1: $C_1 = 50 \mu F$, $R_{11} = 9 \Omega$, $R_{12} = 150 \Omega$, $L_1 = 0.6$ H.	
Load 2: $C_2 = 42 \mu F$, $R_{22} = 5 \Omega$, $R_{12} = 150 \Omega$, $L_2 = 0.4$ H.	
Load 3: $C_3 = 33 \mu F$, $R_{33} = 20 \Omega$, $R_{12} = 150 \Omega$, $L_3 = 1.5$ H.	
Transmission Line data	
TL1: $R_{TL1} = 0.7 \Omega$, $L_{TL1} = 0.5$ mH.	
TL2: $R_{TL2} = 1.5 \Omega$, $L_{TL2} = 0.9$ mH.	
Filter parameters	
$R_f = 1.5$ m Ω , $X_f = 0.5$ mH	
Grid data	
$V = 13.8$ KV, $R_g = 0.2 \Omega$, $L_g = 0.3$ mH	

**FIGURE 1.** SLD of MG.

III. CONTROL STRATEGY

In this paper, the conventional cascaded control system is utilized in maintaining the voltage at the point of common coupling (PCC) for each DG. The reference voltages ($V_{conv_a}^*$, $V_{conv_b}^*$, $V_{conv_c}^*$) are extracted by the transformation of the dq reference voltages ($V_{conv_d}^*$, $V_{conv_q}^*$) using the transformation angle (θ_{PLL}). $V_{conv_d}^*$ and $V_{conv_q}^*$ are extracted with the help of four PI controllers as shown in Fig. 2. θ_{PLL} is excluded from the phase-locked loop (PLL), which sets the grid voltages at its entry level. The firing pulses of the inverter switches are obtained from the comparator, which compares the reference voltages ($V_{conv_a}^*$, $V_{conv_b}^*$, $V_{conv_c}^*$) with a triangular signal. V_d and V_q are the direct and the

**FIGURE 2.** Conventional control technique for autonomous operation mode.

quadrature axis voltages extracted by the transformation from the stationary frame to the dq frame.

The four PI controllers' gains are extracted by the aid of the proposed adaptive technique and the other optimization techniques. It is more stated in details in the Section IV.

IV. MODELING STAGE

A. THE RSM

Thanks to its performance in analysing many systems, the value of the RSM technique is quickly increasing. The RSM is a mathematical technique. It empirically develops models by extracting association between control and reaction parameters by means of a good statistical technique [28]. Through the data collected, the PSCAD program makes a great contribution to the simulation stage. The data extracted from the PSCAD is used as input of the RSM system for the different scenarios. For the introduced research, the RSM input data is decided to be the settling time (T_{set}), maximum percentage undershoots (MPUS), steady-state error (E_{ss}) and maximum percentage overshoots (MPOS) of the terminal voltage. The central composite design (CCD) scheme is utilized to run the 2nd order RSM model for more reliable performance. The RSM is applied by the MINITAB software.

The minimization of the MPOS, MPUS, T_{set} , and E_{ss} for the three scenarios describe the multi-objective function. Eq. (1) is the objective function, which is extracted from the RSM model for scenario 1, which is used in the optimization process for both SFO and PSO.

$$X_i = Z_1 + Z_2k_1 + Z_3k_2 + Z_4k_3 + Z_5k_4 + Z_6k_1^2 + Z_7k_2^2 + Z_8k_3^2 + Z_9k_4^2 + Z_{10}k_1k_2 + Z_{11}k_1k_3 + Z_{12}k_1k_4 + Z_{13}k_2k_3 + Z_{14}k_2k_4 + Z_{15}k_3k_4 \quad (1)$$

where $i = 1, 2, 3, 4$,

$Z_1, Z_2, \dots, Z_{15}$ = the RSM model constants obtained for each scenario which are presented in [29].

B. THE SFO ALGORITHM

The development of soft computing power is the major motivation to use SFO in optimizing many problems. The SFO is a heuristic technique that is motivated by nature. Its core idea is mainly simulating the orientation of the sunflowers

to receive sunlight [9]. The sunflower motion is repeated every day, which starts in the morning following the sunlight. At the evening, the sunflower returns to the initial position waiting for the sunlight. Just one pollen gamete is presumed by each sunflower. Radiation from inverse square rule here is essential. Since the sunflowers take next to the sun a massive amount of heat from the sun compared to those farther ones that of the distant ones. The sunflowers are next to the sun lean towards quiet in this spot [9]. Eq. (2) shows the heat absorbed for each population.

$$Q_i = \frac{P}{4\pi r_i^2} \quad (2)$$

where P is known as the source of the power. r_i is the space between the recent best and population i .

Motions of sunflowers is described by Eq. (3) [10]:

$$\vec{s}_i = \frac{X^* - X_i}{||X^* - X_i||}, \quad i = 1, 2, \dots, n_p. \quad (3)$$

Eq. (4) shows the sunflowers' step in the direction of "s":

$$d_i = T \times P_i(X_i + X_{i-1}) \times ||X_i + X_{i-1}|| \quad (4)$$

where T is a fixed value, which describes an "inertial" movement of the sunflowers, $P_i(||X_i + X_{i-1}||)$ is the pollination probability.

The restriction on these steps is stated in Eq. (5):

$$d_{\max} = \frac{||X_{\max} - X_{\min}||}{2 \times N_{\text{pop}}}, \quad (5)$$

where $X_{\max}$ and $X_{\min}$ are defined as the upper and lower bounds. N_{pop} shows the total number of populations.

The next plant is given by this eq.:

$$\vec{X}_{i+1} = \vec{X}_i + d_i \times \vec{s}_i \quad (6)$$

The SFO steps are stated as:

- 1) Generation of random population.
- 2) The best population is assumed to be the new sun.
- 3) The other populations adjust their orientation towards the sun.

A flowchart for SFO, is shown in Fig. 3 for further explanations.

C. OPTIMIZATION USING SFO

The multi-objective function is stated in Eq. (1). It relies on the weighting method [30], and it is used in the SFO code which is presented in [31]. The settings of the SFO and the weights used in the multi objective function are stated in [29].

D. LMSRE ALGORITHM

The adaptive filtering algorithms (AFA) are typically used to find the impulse response weight vector (W_0) filter [32] as presented in Fig. 4. The input C_k is applied to a Gaussian noise N_k passing through FIR filter. Therefore, it relies on the error e_k , which is presented in Eq. (7) [32].

$$e_k = d_k - W_k^T C_k \quad (7)$$

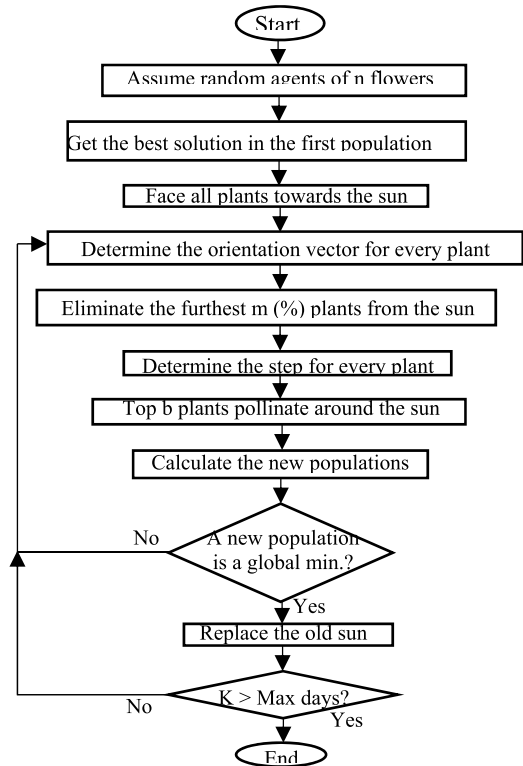


FIGURE 3. SFO algorithm.

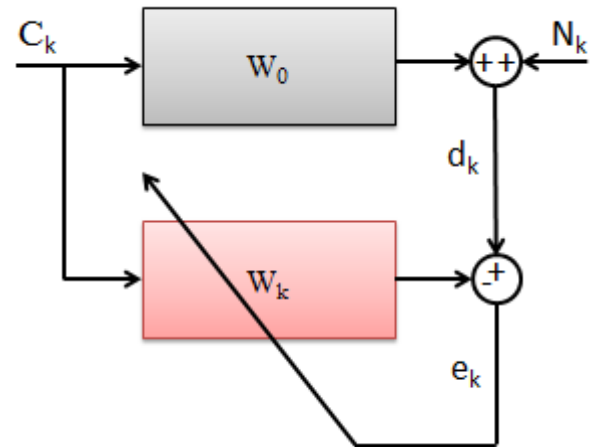


FIGURE 4. System identification model of FIR filter.

where k is the number of iterations, W_k defines the expected vector of the weight. d_k defines the forecasted signal as in Eq. (8)

$$d_k = W_0^T C_k + N_k \quad (8)$$

The iteration of the AFAs relies on the steepest descent algorithm as in Eq. (9).

$$W_{k+1} = W_k - \mu \nabla_w j(W_k) \quad (9)$$

where μ shows the step-size. ∇ is the grad. operative. J defines the cost function, as presented in Eq. (10).

$$j(W_k) = |e_k| + 2\sqrt{1 + \exp(-|e_k|)} - 2\sqrt{2} \quad (10)$$

Then, the gradient of the cost function is obtained as follows:

$$\nabla_{W_j}(W_k) = \frac{\partial}{\partial W_k} |e_k| + 2 \frac{\partial}{\partial W_k} \times \sqrt{1 + \exp(-|e_k|)} \quad (11)$$

$$\frac{\partial}{\partial W_k} |e_k| = \frac{\partial}{\partial W_k} |d_k - W_k^T C| = \text{sign}(e_k) \cdot (-C_k) \quad (12)$$

$$\frac{\partial}{\partial W_k} \sqrt{1 + \exp(-|e_k|)} = \frac{-\exp(-|e_k|) \frac{\partial}{\partial W_k} |e_k|}{2\sqrt{1 + \exp(-|e_k|)}} \quad (13)$$

Then, substitute Eq. (12) and Eq. (13) into Eq. (11) to get Eq. (14) and Eq. (15).

$$\begin{aligned} \nabla_{W_j}(W_k) &= \left[\text{sign}(e_k) \cdot (-C_k) - \frac{\exp(-|e_k|) \text{sign}(e_k) \cdot (-C_k)}{\sqrt{1 + \exp(-|e_k|)}} \right] \end{aligned} \quad (14)$$

$$\begin{aligned} \nabla_{W_j}(W_k) &= \text{sign}(e_k) \cdot (-C_k) - \left[\frac{\exp(-|e_k|)}{\sqrt{1 + \exp(-|e_k|)}} \right] \end{aligned} \quad (15)$$

Assume that

$$\beta_k = 1 - \frac{\exp(-|e_k|)}{\sqrt{1 + \exp(-|e_k|)}} \quad (16)$$

Then, substitute Eq. (15) into Eq. (9) to get Eq. (17).

$$W_{k+1} = W_k - \mu_k \cdot \beta_k \cdot \text{sign}(e_k) \cdot C_k \quad (17)$$

where μ_k is to differ with the variance of the error, for the huge error, μ_k should be large for rapid convergence. On the other hand, for a small error, μ_k must be minimized. So, β_k varies from [0, 1], which is minimized for tiny error and vice versa. So, μ_k varies proportionally to the β_k as below.

$$\mu_k = \mu \cdot \beta_k^{\alpha-1} \quad (18)$$

where μ and α are responsible for variation of μ_k , then, substitute Eq. (18) into Eq. (17) to obtain eq. Eq. (19).

$$W_{k+1} = W_k - \mu \cdot \beta_k^\alpha \cdot \text{sign}(e_k) \cdot C_k \quad (19)$$

E. OPTIMIZATION USING LMSRE

The PI Controller Parameters are modified using the LMSRE algorithm Based on eq. (19). The updated PI parameters can be defined below:

$$k_{p(k+1)} = k_{p(k)} + \Delta k_{p(k)} \quad (20)$$

$$T_{i(k+1)} = T_{i(k)} + \Delta T_{i(k)} \quad (21)$$

$$\Delta k_{p(k)} = \Delta T_{i(k)} = \mu \cdot \beta_k^\alpha \cdot \text{sign}(e_k) \cdot C_k \quad (22)$$

The initial PI parameter values (k_p and T_i) for the 6 PI controllers (V_1 to V_6) are obtained with the trial-and-error method expressed in Table 2.

TABLE 2. The initial PI parameter values for scenario 1.

controller	V_1	V_2	V_3	V_4	V_5	V_6
Initial k_p	5	3	5	3	5	3
Initial T_i	0.003	0.3	0.003	0.3	0.003	0.3

V. COMPARATIVE EVALUATION FOR THE SFO, PSO AND LMSRE RESULTS

This section's primary goal is to proof the validity and to display the effectiveness of the suggested control strategy to sustain the PCC voltage within the required limits in various MG operating conditions. The proposed practicality of the control plan is shown by the data of simulation which is extracted from PSCAD/EMTDC software. The results of SFO and PSO in [29] are compared with the obtained from LMSRE, the constants of the LMSRE are assumed to be ($\mu = 0.01$), and ($\alpha = 5.5$). For more accuracy, 50 μs is selected for both time step and the channel plot time. The system is evaluated under several different environments which are: 1) Disconnection of the system from the grid (islanding), 2) Autonomous system exposure to load variability, and 3) Autonomous system exposure to a symmetrical fault. More detailed discussion is presented in the next sections.

TABLE 3. The results of LMSRE, SFO and PSO for Scenario1 in the 3 DGs.

Point Of comparison	LMSRE	SFO	PSO
Scenario 1 DG 1			
MPUS	7.9%	12.905%	20.26%
Tset	0.0461 s	0.033 s	0.054 s
Ess	0.32%	0.35%	0.4%
Scenario 1 DG 2			
MPUS	7.898%	12.86%	20.1%
Tset	0.0461 s	0.0326 s	0.0535 s
Ess	0.3%	0.36%	0.405%
Scenario 1 DG 3			
MPUS	7.57%	12.6%	19.975%
Tset	0.0432 s	0.0322 s	0.0531 s
Ess	0.31%	0.32%	0.402%

A. SCENARIO 1 (AUTONOMOUS OPERATION)

In this Scenario, the MG is in a steady-state condition in the grid-connected mode. The MG is suddenly disconnected from the utility grid (autonomous mode) at $t = 2$ s. The optimal PI parameters values in each DG for LMSRE, SFO and PSO are listed in Table 3. Figs. 5 (a, b, c) present the PCC voltage profile in each DG for LMSRE, SFO and PSO. Figs. 6 (a, b, c) present the Real and imaginary powers for the load in each DG for LMSRE, SFO and PSO. It is valuable

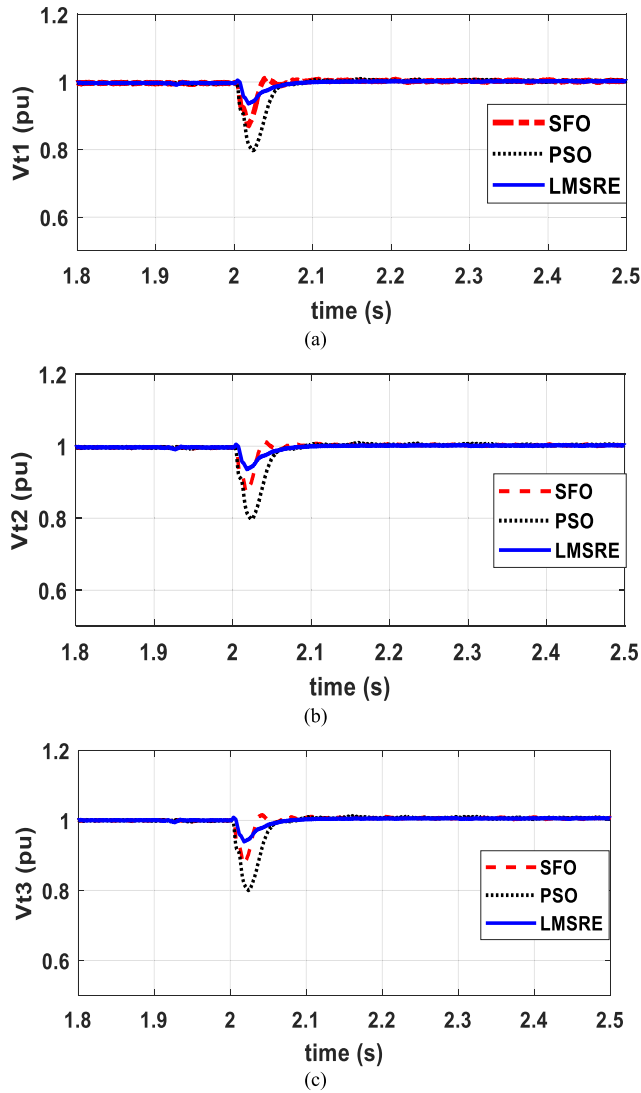


FIGURE 5. Output results for LMSRE, SFO and PSO of scenario 1. (a) Voltage profile of DG 1. (b) Voltage profile of DG 2. (c) Voltage profile of DG 3.

to mention that, in Fig. 5a, the MPUS for autonomous mode responses of the PCC1 voltage for the proposed technique are less than 8%. Moreover, the settling time based on 2% criterion is 5 ms and the E_{ss} is 0.32%. Therefore, the proposed control technique shows minimum overshoots, rapid damping and perfect E_{ss} . The voltage undershoot of the DG 3 is slightly lower than other DGs owing to its higher distance from the fault place, as seen in Fig. 5c. It is valuable to mention that the LMSRE is pioneer in MPUS, MPOS, test, and is compared to SFO and PSO. This proves the flexibility, justification, and applicability of the introduced adaptive technique (LMSRE) over SFO and PSO.

B. SCENARIO 2 (LOAD CHANGING)

In this Scenario, the MG is run in the islanded mode. The MG originally runs with the RLC loads, which is mentioned in Table 1. R12 is changed from 150 Ω to 300 Ω

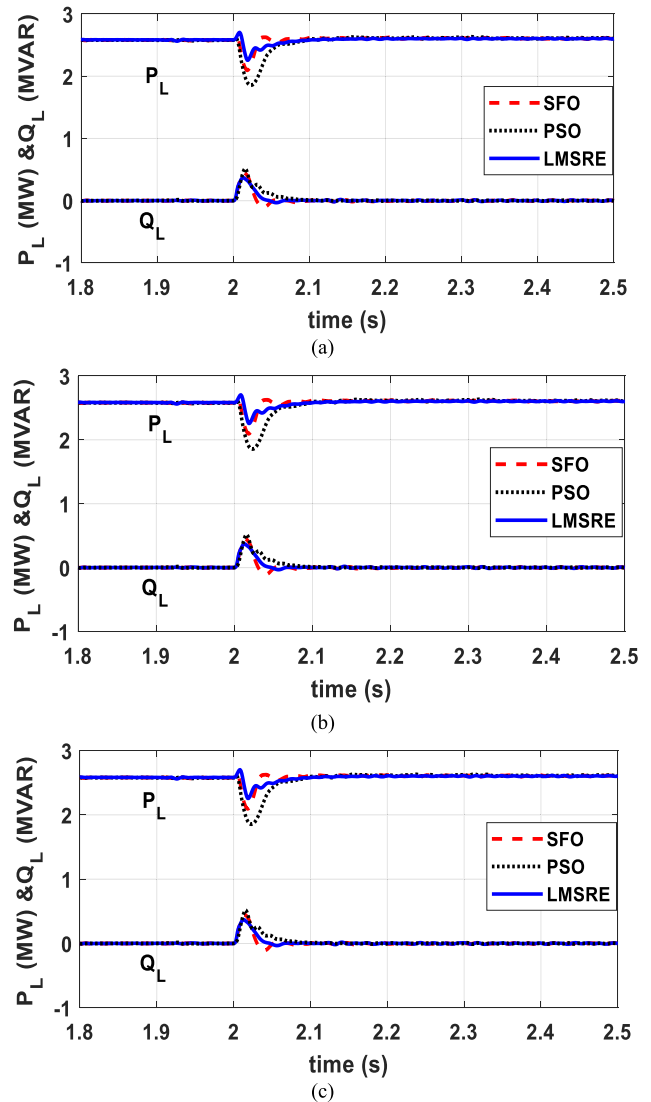
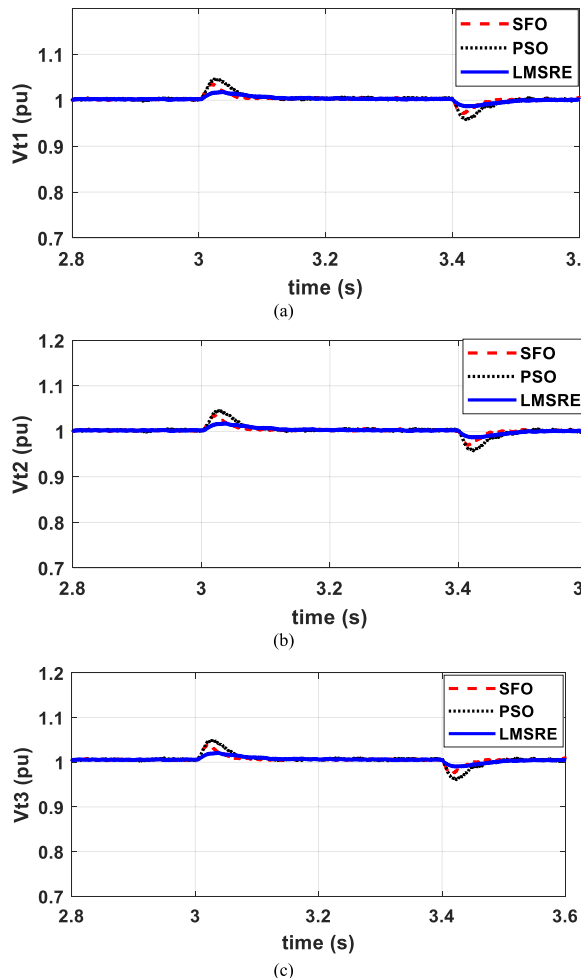


FIGURE 6. Output results for LMSRE, SFO and PSO of scenario 1. (a) Real and reactive powers for the load in DG 1. (b) Real and imaginary powers for the load in DG 2. (c) Real and imaginary powers for the load in DG 3.

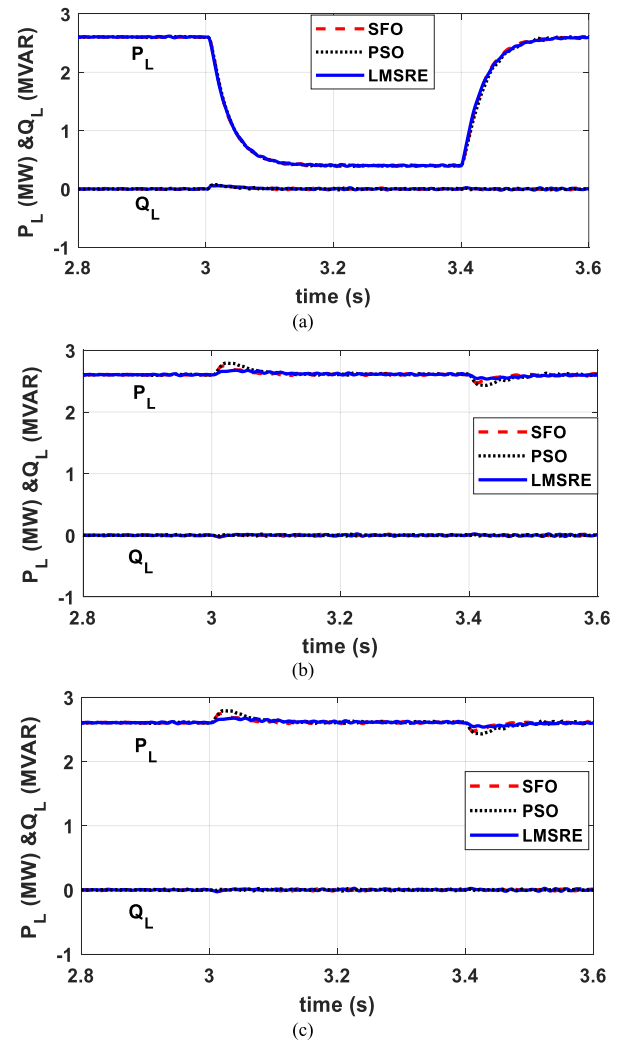
at $t = 3$ s and is returned to its initial state at $t = 3.4$ s. The optimal PI parameters values in each DG for LMSRE, SFO and PSO are listed in Table 4. Figs. 7 (a, b, c) present the PCC voltage profile in each DG for LMSRE, SFO and PSO. Figs. 8 (a, b, c) present the Real and imaginary powers for the load in each DG for LMSRE, SFO and PSO. It is valuable to mention that, in Fig. 7a, the MPUS and MPOS for the load changing responses of the PCC1 voltage for the proposed technique are less than 3%. Moreover, the settling time based on 2% criterion is 40 ms and the E_{ss} is 0.43%. Therefore, the proposed control technique shows minimum overshoots, rapid damping and perfect E_{ss} . The voltage overshoots of the DG 3 are slightly lower than other DGs owing to its higher distance from the disturbance place, as seen in Fig. 7c. It is valuable to realize that, in Fig. 8a, the active power of DG 1 load decreased from 2.6 MW to 0.5 MW and back to its initial state smoothly at $t = 3.4$ s.

TABLE 4. The results of LMSRE, SFO and PSO for Scenario2 in the 3 DGs.

Point Of comparison	LMSRE	SFO	PSO
Scenario 2 DG 1			
MPUS	1.88%	3.2%	4.3%
MPOS	2.22%	3.97%	5%
Tset	0.401 s	0.43 s	0.45 s
Ess	0.43%	0.44%	0.48%
Scenario 2 DG 2			
MPUS	1.882%	3.12%	4.275%
MPOS	2.1%	3.61%	4.64%
Tset	0.403 s	0.4289 s	0.445 s
Ess	0.52%	0.665%	0.69%
Scenario 2 DG 3			
MPUS	1.5%	2.745%	3.909%
MPOS	2.53%	3.6%	4.7%
Tset	0.4015 s	0.424 s	0.443 s
Ess	0.737%	1.032%	1.07%

**FIGURE 7.** Output results for LMSRE, SFO and PSO of scenario 2. (a) Voltage profile of DG 1. (b) Voltage profile of DG 2. (c) Voltage profile of DG 3.

On the other hand, the active powers for the other DG loads have rapid damping with smaller fluctuations. It is valuable

**FIGURE 8.** Output results for LMSRE, SFO and PSO of scenario 2. (a) Real and imaginary powers for the load in DG 1. (b) Real and imaginary powers for the load in DG 2. (c) Real and imaginary powers for the load in DG 3.

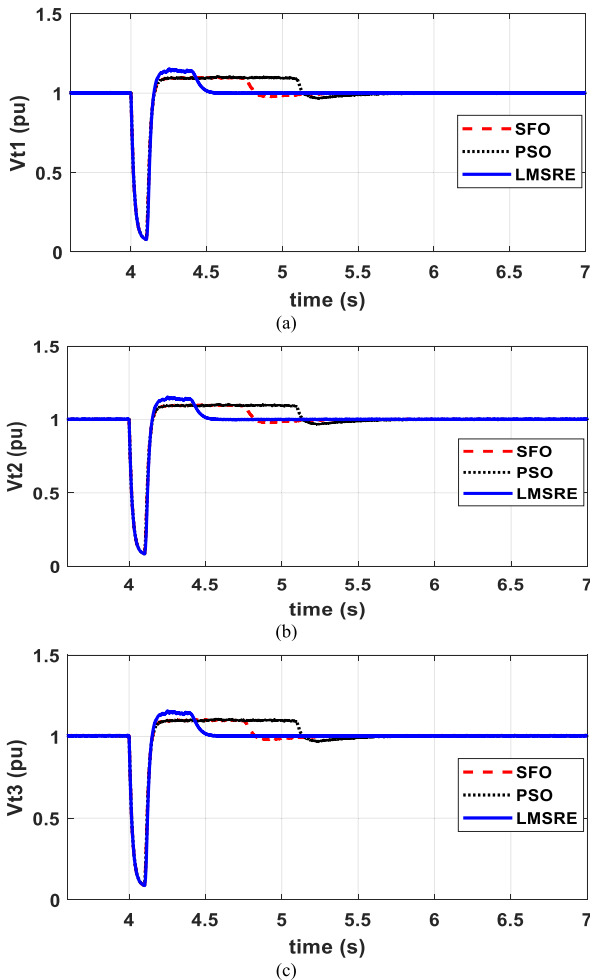
to mention that the LMSRE pioneer in MPUS, MPOS, T_{set} , and E_{ss} compared to SFO and PSO, which proof the flexibility, justification, and applicability of the introduced adaptive technique (LMSRE) over SFO and PSO.

C. SCENARIO 3 (SYMMETRICAL FAULT)

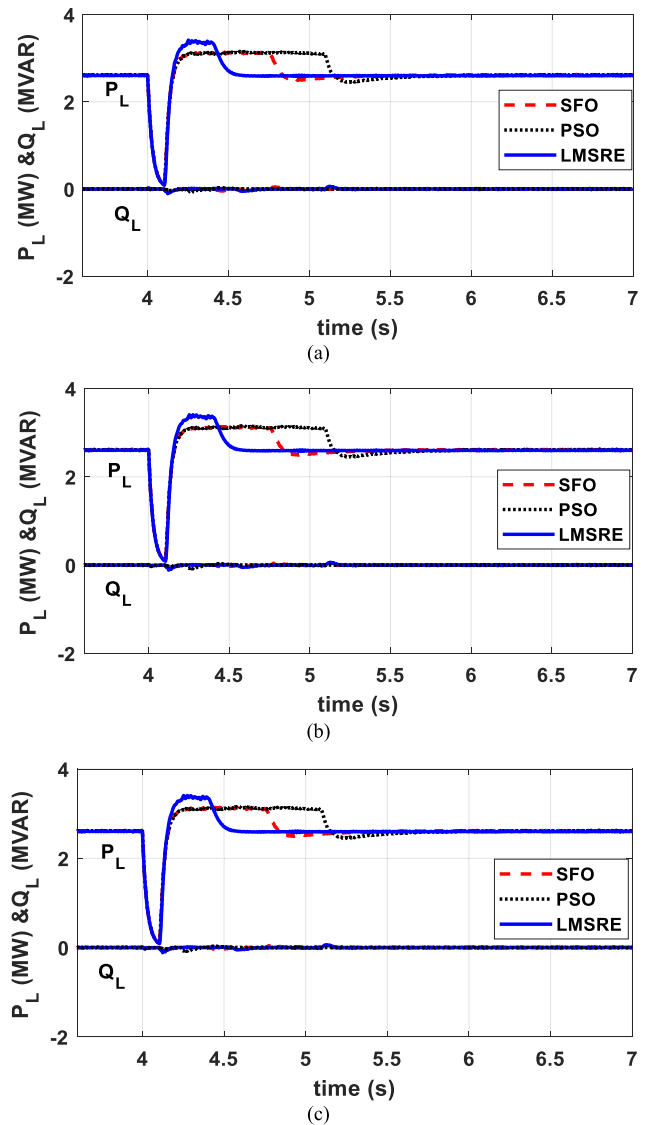
In this Scenario, the MG is at steady-state in an islanded mode. Then, a symmetrical fault is suddenly occurs at PCC 1 at $t = 4$ s and back to a healthy state at $t = 4.1$ s. The optimal PI parameters values in each DG for LMSRE, SFO and PSO are listed in Table 5. Figs. 9 (a, b, c) present the PCC voltage profile in each DG for LMSRE, SFO and PSO. Figs. 10 (a, b, c) present the real and imaginary powers for the load in each DG for LMSRE, SFO and PSO. It is valuable to mention that, in Fig. 9a, the T_{set} based on 2% criterion for the fault responses of the PCC1 Voltage for the proposed technique is 49 ms and the E_{ss} is 0.25%. Therefore, the proposed control technique shows rapid damping and perfect E_{ss} . The voltage overshoots of the DG 3 are slightly lower

TABLE 5. The results of LMSRE, SFO and PSO for Scenario3 in the 3 DGs.

Point Of comparison	LMSRE	SFO	PSO
Scenario 3 DG 1			
MPUS	92.16%	91.638%	93.108%
MPOS	15.4%	10.705%	10.956%
Tset	0.4915 s	0.9654 s	1.3901 s
Ess	0.25%	0.46%	0.55%
Scenario 3 DG 2			
MPUS	91.67%	92.135%	92.14%
MPOS	15.402%	10.3505%	10.604%
Tset	0.4916 s	0.9654 s	1.3901 s
Ess	0.337%	0.495%	0.63%
Scenario 3 DG 3			
MPUS	91.312%	93.106%	91.64%
MPOS	15.824%	10.205%	10.59%
Tset	0.4972 s	0.8069 s	1.384 s
Ess	0.67%	0.842%	1.02%

**FIGURE 9.** Output results for LMSRE, SFO and PSO of Scenario 3. (a) Voltage profile of DG 1. (b) Voltage profile of DG 2. (c) Voltage profile of DG 3.

than other DGs owing to its higher distance from the fault place, as seen in Fig. 9c. It is valuable to mention that the

**FIGURE 10.** Output results for LMSRE, SFO and PSO of scenario 3. (a) Real and imaginary powers for the load in DG 1. (b) Real and imaginary powers for the load in DG 2. (c) Real and imaginary powers for the load in DG 3.

LMSRE has slightly higher overshoots. On the other hand, it shows very fast damping and perfect E_{ss} compared to SFO and PSO. This proves the flexibility, justification, and applicability of the introduced adaptive technique (LMSRE) over SFO and PSO.

VI. CONCLUSION

This paper has introduced a new implementation of adaptive PI controller optimum design scheme of multiple PI controllers' parameters to boost microgrid efficiency. Six PI controllers are utilized in the control scheme. The results showed that the presented controller is able to simultaneously retain the real and imaginary power. It also regulates the reference voltages. The results show a quick damping in transient response with a rapid T_{set} and a small E_{ss} under a different microgrid operating condition. The system is evaluated

under several different environments which are: 1) disconnect the system from the grid (islanding), 2) autonomous system exposure to load variability, and 3) autonomous system exposure to a symmetrical fault. The proposed practicality of the control plan is shown by the data of the simulation which is extracted from PSCAD/EMTDC software. The strength of the suggested adaptive control is confirmed through matching its results with that obtained using the SFO and the PSO techniques. LMSRE-based adaptive PI controller has achieved lower values of the transient responses in the autonomous operation of the MG over that obtained using the SFO and the PSO. In scenario 1, The MPUS of the voltage profile using the suggested adaptive control is decreased by 38.7% and 61% over that using the SFO and the PSO, respectively. In scenario 2, The MPUS of the voltage profile using the suggested adaptive control is decreased by 41% and 56% over that using the SFO and the PSO, respectively. Moreover, the MPOS of the voltage profile using the LMSRE is decreased by 44% and 55.6% over that by using the SFO and the PSO, respectively. In scenario 3, The T_{set} of the voltage profile using the suggested adaptive control is decreased by 49% and 64% over that using the SFO and the PSO. Furthermore, the E_{ss} of the voltage profile using the LMSRE is decreased by 45.6% and 54.5% over that using the SFO and the PSO, respectively. The future work will focus on intensifying the proposed LMSRE-based adaptive PI controller to adjust the power system applications, energy storage devices, and smart grids, achieving optimum responses in the renewable energy conversion system.

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